

NorthBridge Distribution

Supply Chain Planning & Inventory Risk Analysis

Latest snapshot: 31 August 2026

1. Background and Overview

NorthBridge Distribution operates a multi-warehouse FMCG distribution network managing 180 SKUs across 18 suppliers and 6 warehouse locations. The analysis focuses on inventory availability, forecast demand, supplier lead times and replenishment risk across the network.

The business question for Q1 is: Which SKUs are projected to stock out before replenishment can arrive, how many units are at risk, and what potential revenue is exposed?

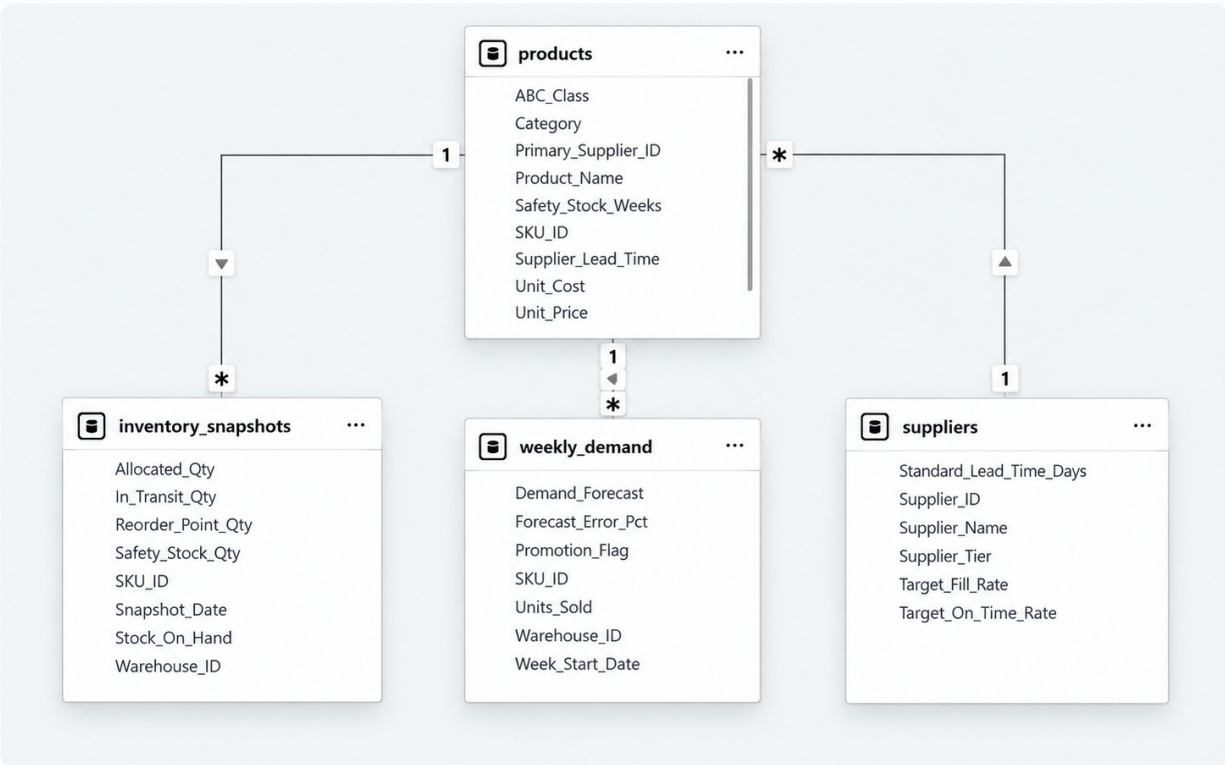
The objective is to identify stockout exposure early enough for planners to prioritise intervention, rather than waiting until stock reaches zero.

2. Data Structure Overview

Q1 combines product, inventory, demand and supplier information. Products acts as the shared SKU dimension linking inventory and weekly demand, while each product is associated with its primary supplier.

Table	Key field	Main fields used	Purpose
products	SKU_ID	Primary_Supplier_ID, Unit_Price	SKU dimension and commercial value
inventory_snapshots	SKU_ID	Snapshot_Date, Stock_On_Hand, Allocated_Qty	Current available inventory
weekly_demand	SKU_ID	Week_Start_Date, Demand_Forecast	Expected weekly demand
suppliers	Supplier_ID	Standard_Lead_Time_Days	Expected replenishment time

Data Model



3. Executive Summary

52 SKUs at risk	17,408 Projected shortage units	£1,168,343 Potential revenue at risk	SKU-1092 Largest shortage: 1,167 units
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Using the latest inventory and weekly-demand snapshot, 52 of 180 SKUs have fewer weeks of cover than the time required for their primary supplier to replenish them. These SKUs represent approximately 17,408 projected shortage units and £1,168,343 of potential revenue at risk if forecast demand cannot be fulfilled before replenishment arrives.

SKU-1092 has the largest projected unit shortage at 1,167 units, while SKU-1027 carries the highest potential revenue exposure at approximately £75,922. This means replenishment priorities should consider both operational shortage volume and financial exposure.

4. Insights Deep Dive

Stockout risk affects a meaningful share of the portfolio. 52 of 180 SKUs are projected to run out before normal supplier replenishment can arrive, meaning roughly 29% of the SKU portfolio requires attention.

The network faces both operational and financial exposure. The risk SKUs represent approximately 17,408 units of projected shortage and around £1.17M of potential revenue exposure.

SKU-1092 is the largest volume-risk priority. With approximately 1,167 projected shortage units, SKU-1092 creates the greatest availability and fulfilment pressure by unit volume.

The largest unit shortage is not necessarily the greatest financial risk. SKU-1027 has the highest potential revenue exposure at approximately £75.9K, showing why prioritisation should not rely on shortage quantity alone.

Supplier lead time materially changes the risk assessment. The analysis compares Weeks of Cover with replenishment lead time. A SKU can still have stock remaining and be at risk when that stock will not last until replenishment arrives.

Revenue at risk is an exposure estimate. The approximately £1.17M represents potential sales exposure if forecast demand cannot be fulfilled; mitigation such as transfers, expediting, substitution or partial fulfilment may reduce the realised impact.

Priority risk examples:

SKU	WOC	Lead time (weeks)	Projected shortage	Revenue at risk
SKU-1092	2.32	4.00	1,167	£65,889
SKU-1003	2.34	4.00	918	£57,944
SKU-1033	2.82	4.00	632	£48,854
SKU-1069	2.90	4.00	626	£56,465
SKU-1027	2.88	4.00	624	£75,922

5. Recommendations

- **Create a risk-based replenishment priority.** Rank the 52 risk SKUs using both projected shortage quantity and potential revenue at risk rather than treating every warning equally.
- **Escalate high-value and high-volume risks separately.** Prioritise financially exposed SKUs such as SKU-1027 while separately managing high-volume risks such as SKU-1092.
- **Trigger replenishment earlier for vulnerable SKUs.** Where Weeks of Cover repeatedly approaches or falls below supplier lead time, review reorder points and safety-stock settings so purchasing is triggered earlier.
- **Check internal inventory before raising a new purchase order.** Assess whether shortage SKUs have excess inventory at another warehouse and rebalance stock where practical before increasing total network inventory.
- **Use exception-based monitoring.** Maintain a Power BI view that flags WOC below supplier lead time and ranks exceptions by shortage quantity and revenue exposure.
- **Investigate recurring risk at root-cause level.** For SKUs that repeatedly appear, review forecast error, supplier lead time, safety stock, allocation policy and purchasing timing rather than relying on repeated expediting.

Management decision supported: Q1 converts inventory, demand and supplier lead-time data into a prioritised replenishment decision: which SKUs require intervention first, how large the expected shortage is, and how much potential revenue is exposed.

